

# Tejus Singh

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## Education

### University of California, San Diego

Expected Dec 2026

Masters of Science in Computer Science | GPA: 4.00

### University of California, Irvine

Sep 2022 - Jun 2025

Bachelors of Science in Applied Physics, Minor in Mathematics | GPA: 3.80

Relevant Courses: LLM theory & reasoning, Algorithms (DSA), Embedded systems, OS, Signal processing, Particle physics, Time series modeling, Quantitative finance & derivatives, Linear & abstract algebra

## Technical Skills

**Languages and Tools:** Python, Julia, C/C++, Java, SQL, TensorFlow, Pytorch, JAX, Git, Pandas, Vitis HLS, CUDA

**Skills:** Probabilistic and online machine learning, Predictive modeling, Quantitative trading, Portfolio management, Computational medical imaging, Software development, Embedded systems software

## Experience

### Kastner Lab | Graduate Researcher - La Jolla, CA

Nov 2025 - April 2026

- Implementing **quantized transformer architecture** for **sub-microsecond inference** on AMD Versal FPGA
- Compressing a full transformer (8 Layer/8 Heads) into a hardware-compatible (4 layer/4 head) model, through architecture simplification and retraining in **C++** and **Python**, without degrading accuracy (0.86 in validation)
- Working with CERN and Fermilab on **particle classification** for real-time physics inference

### Evovest | Quantitative Research Intern - Montréal, Canada

Mar 2024 - Jun 2025

- Researched, produced asset returns prediction models using **Bayesian regime-switching Kalman filters**, **variational autoencoding**, **physics-informed loss** functions, and **neural stochastic differential equations**.
- Outperformed Gradient Boosted Trees (**XGBOOST**) and flavors of VAR and (S)ARIMA models
- Built **trust-aware prediction systems** by fusing ML model outputs with past and incoming market signals using **bayesian uncertainty** to stabilize performance during high-volatility regimes
- Owned full project development from **research to production**, handling large tabular data and streaming pipelines

### Molloi Imaging Physics Lab | Undergraduate Researcher - Irvine, CA

Jan 2023 - June 2025

- Published voxel-wise deep-learning pipelines for medical imaging, 3D CT segmentation, and material decomposition.
- Implemented **physics-based constraints with neural networks** to improve learning and generalization
- Ran large scale 3D simulations and experiments on patient data to benchmark model performance
- Researched **CNNs**, **UNets**, and **autoencoders** in **Julia** and **Python**, leveraging **CUDA** for acceleration

## Publications

D. Black, T. Singh, S. Molloi. **Coronary artery calcium quantification using dual energy material decomposition: A simulation study**. Int. J. Cardiovascular Imaging, vol. 40, no. 7, 1465–1474, 2024

<https://doi.org/10.1007/s10554-024-03124-9>

## Projects

### Time Series Forecasting via Transformers

- Forecasting volatile signals via the **inverted transformer**. **Feature engineering** and extracting low-dimension structures via unsupervised methods (**manifold learning**, autoencoding, **wavelet analysis**)
- Building **AI Agent** pipeline to aggregate and automate analysis of non-numerical, multi-modal data

### Nitrogen Vacancy Centers - <https://drive.google.com/file/d/1KdeWQuoe9psiV-UMVnv5jwJUMmqf9Zb/view>

- Studied spin-based qubit systems, connected physical spin dynamics to computational primitives, and manipulated electron spin states to access various quantum states

### Creative Problem Solving in Mathematics - <https://drive.google.com/file/d/1FyG6ULg18AhrNRF8esVFyAor-grDAFjK/view>

- Compilation of the ways to prove ideas mathematically. Written with inspiration from Polya's "How to solve it," under guidance of mathematician Jaaziel Lopez de la Luz

### Robotic Balancing via Control Algorithms

- CAD designed, 3D printed, and programmed an arduino platform robot to balance and move a ball to an arbitrary position